

Abstract

The **Agentic AI-Based Travel Planning Assistant Using LangChain** is an intelligent AI-powered travel recommendation and itinerary generation system developed using Python, LangChain, Streamlit, and Large Language Models (LLMs). The application automates the process of trip planning by integrating flight search, hotel recommendation, tourist attraction discovery, weather forecasting, budget estimation, and itinerary generation into a single intelligent platform.

Traditional travel planning requires users to manually browse multiple websites, compare information, calculate budgets, and organize schedules. This project solves that problem using an Agentic AI workflow where autonomous LangChain agents intelligently select tools, analyze travel datasets, fetch live weather information, and generate optimized travel plans. The system uses real JSON datasets (flights.json, hotels.json, and places.json) along with the Open-Meteo weather API to provide realistic and personalized travel recommendations. The application also includes conversational AI chat support, multi-city planning, downloadable reports, recommendation explanations, and a modern Streamlit-based user interface. This project demonstrates the practical implementation of Agentic AI systems in the travel and tourism domain using modern LLM orchestration frameworks.

1: Introduction

1.1 Introduction

Travel planning is often a time-consuming and complex process that requires users to compare flights, hotels, attractions, weather conditions, and budgets across multiple platforms. Most travelers manually gather information from different websites, which may lead to inefficient planning and poor decision-making.

With advancements in Artificial Intelligence, conversational agents and autonomous AI systems can now simplify travel planning by automatically analyzing user requirements and generating optimized itineraries.

Agentic AI-Based Travel Planning Assistant Using LangChain is designed to automate and enhance the travel planning process using:

- LangChain tool-calling agents
- Real travel datasets
- Live weather APIs
- AI-powered recommendations
- Intelligent itinerary generation

The system acts as a virtual AI travel assistant capable of understanding user preferences and generating structured travel plans in real time.

1.2 Problem Statement

Planning a trip involves multiple challenges:

- Searching flights and hotels separately
- Comparing inconsistent information
- Managing travel budgets manually
- Checking weather forecasts
- Creating efficient itineraries
- Filtering attractions based on preferences

Users often spend significant time switching between multiple travel platforms. There is a need for an intelligent automated system that can:

- Understand travel requirements
 - Analyze multiple travel factors
 - Generate optimized travel plans
 - Provide personalized recommendations
 - Save users time and effort
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1.3 Objectives

Primary Objectives

1. Build an Agentic AI travel planning system using LangChain.
2. Integrate travel datasets and weather APIs.
3. Implement autonomous tool-calling workflows.
4. Generate day-wise travel itineraries.
5. Estimate travel budgets dynamically.

Secondary Objectives

1. Provide conversational AI support.
 2. Implement intelligent recommendation logic.
 3. Add multi-city trip planning.
 4. Support PDF and report exports.
 5. Create a responsive modern user interface.
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1.4 Scope of the Project

The project focuses on:

- Domestic travel planning
- Dataset-based recommendations
- Weather-aware itinerary generation
- Budget estimation
- AI-powered recommendations
- Interactive Streamlit interface

The system can be extended in future for:

- Real-time flight APIs
- International travel support
- Payment integration
- Hotel booking systems

- GPS navigation support
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2: Literature Survey

2.1 Existing Systems

Traditional travel websites such as:

- [MakeMyTrip](#)
- [Booking.com](#)
- [Ixigo](#)

provide travel booking services but require users to manually:

- Compare flights
- Search hotels
- Organize itineraries
- Manage budgets

These systems lack autonomous AI reasoning capabilities.

2.2 Proposed System

The proposed system uses:

- LangChain agents
- Tool-calling workflows
- AI-powered reasoning
- Automated itinerary generation

Advantages:

- Faster planning
 - Intelligent recommendations
 - Automated decision-making
 - Personalized trip generation
-

3: System Analysis

3.1 Functional Requirements

The system should:

- Search flights
 - Recommend hotels
 - Suggest attractions
 - Fetch weather forecasts
 - Estimate budgets
 - Generate itineraries
 - Support AI chat
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3.2 Non-Functional Requirements

Performance

- Fast dataset processing
- Responsive UI

Reliability

- Error handling
- Fallback recommendation system

Usability

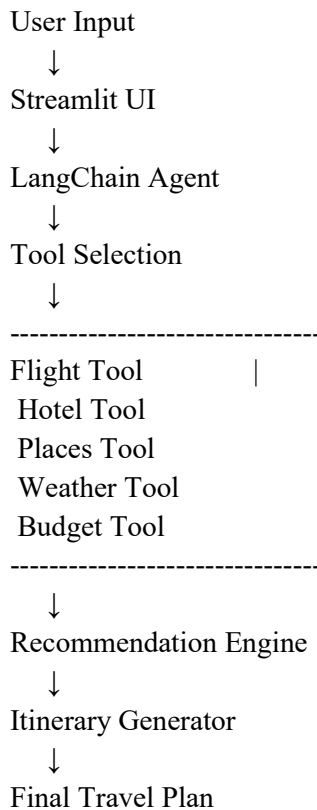
- User-friendly interface
- Interactive dashboard

Security

- Environment variable protection
 - API key security
-

4: System Design

4.1 Architecture Diagram



4.2 Project Structure

```
travel-ai-agent/
├── app.py
├── agents/
├── tools/
├── services/
├── utils/
├── data/
└── output/
```

Chapter 5: Technologies Used

Technology	Purpose
Python	Backend development
LangChain	Agent orchestration
Streamlit	Frontend UI
Gemini/OpenAI	Large Language Model
JSON	Dataset storage
Open-Meteo API	Weather forecasting

6: Modules Description

6.1 Flight Search Tool

Functions:

- Search flights
- Cheapest flight recommendation
- Fastest route suggestion

Dataset:

- flights.json
-

6.2 Hotel Recommendation Tool

Functions:

- Hotel filtering
- Rating-based recommendations
- Amenities filtering

Dataset:

- hotels.json
-

6.3 Places Discovery Tool

Functions:

- Attraction discovery
- Category filtering
- Popular places recommendation

Dataset:

- places.json

6.4 Weather Tool

Uses:

- Open-Meteo API

Provides:

- Current weather
 - Temperature
 - Forecasts
-

6.5 Budget Tool

Calculates:

- Flight cost
 - Hotel cost
 - Food expenses
 - Transport cost
 - Total budget
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7: LangChain Agent Workflow

7.1 Agentic AI Workflow

The LangChain agent:

1. Understands user query
 2. Selects appropriate tools
 3. Retrieves data
 4. Performs reasoning
 5. Generates itinerary
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7.2 Tool Calling

The agent dynamically invokes:

- Flight search tool
- Hotel recommendation tool
- Weather tool
- Places tool
- Budget tool

based on user requests.

8: Implementation

8.1 Streamlit User Interface

The application includes:

- Sidebar inputs
- Dashboard cards
- Weather section

- Budget analysis
 - Itinerary timeline
 - Chat interface
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8.2 Smart Recommendation Engine

The system includes:

- Typo correction
- Intelligent filter relaxation
- Fallback recommendations
- Dynamic dataset normalization

Example:

- “Grdens” → “Gardens”
 - Relaxed hotel budget filtering
-

8.3 Export Features

Users can export travel plans in:

- JSON
 - Markdown
 - Text
 - PDF
-

9: Testing

9.1 Unit Testing

Modules tested:

- Flight tool
 - Hotel tool
 - Weather API
 - Budget calculation
-

9.2 Integration Testing

Tested:

- Agent workflows
 - Tool integration
 - Streamlit interaction
-

9.3 User Interface Testing

Verified:

- Responsive design
 - Dark mode
 - Sidebar interactions
 - Export buttons
-

10: Results and Outputs

Sample Output

Trip Summary

3-day trip from Hyderabad to Delhi.

Flight

Cheapest available flight selected.

Hotel

Budget-friendly hotel with amenities.

Attractions

Top-rated tourist places.

Weather

Live forecast integrated.

Budget

Estimated total trip cost generated.

11: Advantages

- Automated travel planning
 - Intelligent recommendations
 - Time-saving
 - Weather-aware itineraries
 - Conversational AI assistance
 - Modern user experience
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12: Limitations

- Uses local datasets
 - Limited city coverage
 - No real booking integration
 - Dependent on dataset quality
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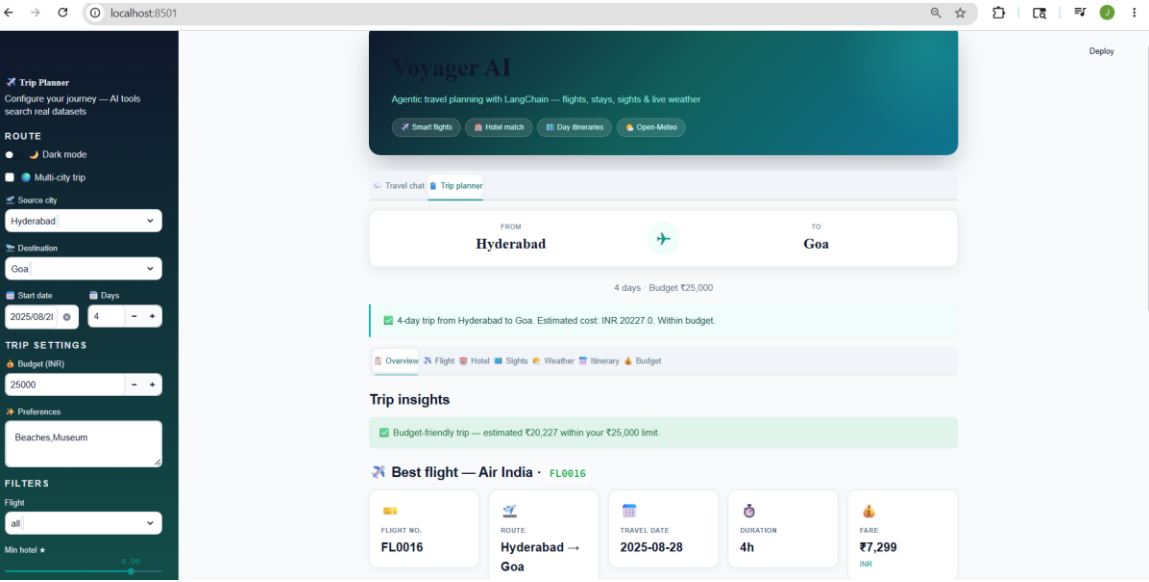
Chapter 13: Future Enhancements

Future improvements may include:

- Real-time flight APIs
 - Hotel booking integration
 - GPS navigation
 - Voice assistant
 - International trip planning
 - AI personalization
 - Mobile application support
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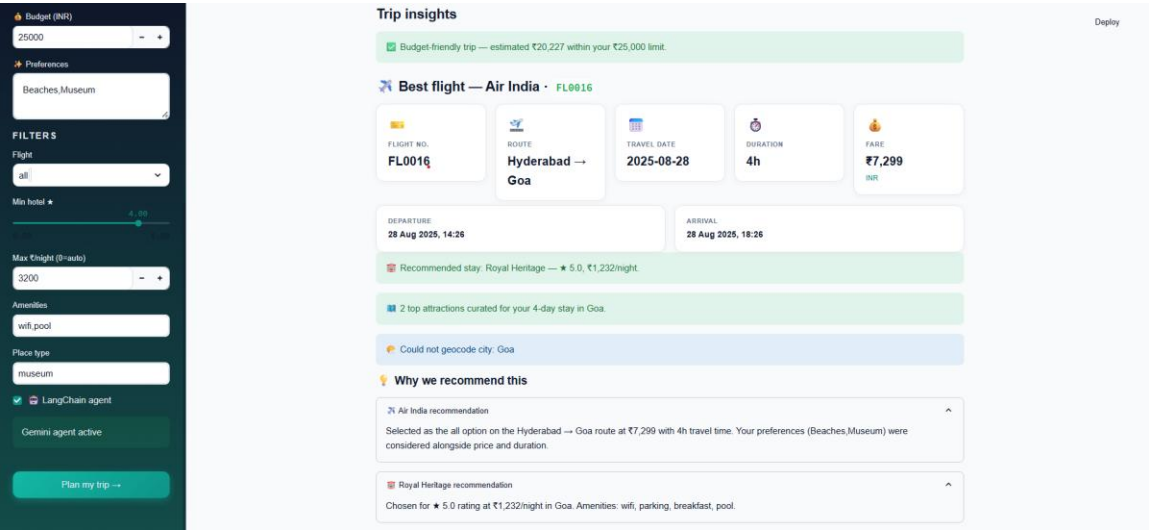
14: Application Screenshots with Description

Home Dashboard



This screen shows the main dashboard where users enter source city, destination, budget, preferences, and travel details.

Recommendation Engine



This screen displays intelligent fallback recommendations for flights, hotels, and attractions when exact matches are unavailable.

Budget (INR)
25000

Preferences
Beaches, Museum

FILTERS

Flight
all

Min hotel
4.00

Max Night (3=auto)
3200

Amenities
wifi, pool

Place type
museum

LangChain agent
Gemini agent active

DEPARTURE
28 Aug 2025, 14:26

ARRIVAL
28 Aug 2025, 18:26

Recommended stay: Royal Heritage — ★ 5.0, ₹1,232/night.

2 top attractions curated for your 4-day stay in Goa.

Could not geocode city: Goa

Why we recommend this

- Air India recommendation**
Selected as the all option on the Hyderabad → Goa route at ₹7,299 with 4h travel time. Your preferences (Beaches, Museum) were considered alongside price and duration.
- Royal Heritage recommendation**
Chosen for ★ 5.0 rating at ₹1,232/night in Goa. Amenities: wifi, parking, breakfast, pool.
- Attractions selection**
Curated 2 top-rated sights in Goa, led by Famous Park, spread across your 4-day itinerary.
- Budget breakdown logic**
Total ₹20,227 includes flight (₹7,299), hotel stays (₹4,928), local transport, and meals for 4 days.

Download your trip

JSON Markdown Text PDF

15: Conclusion

The **Agentic AI-Based Travel Planning Assistant Using LangChain** successfully demonstrates how Agentic AI systems can automate complex travel planning workflows. The project integrates LangChain agents, real travel datasets, live weather APIs, and a modern Streamlit interface to provide intelligent travel recommendations and itinerary generation. The system improves user experience by reducing manual effort, providing personalized recommendations, and enabling autonomous AI-driven travel planning. This project highlights the practical application of LangChain and LLM orchestration in the travel and tourism industry.

References

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